Multi-Site FortiGate Deployment

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1. Executive Summary

1.1 Document Overview

This document provides comprehensive documentation for the EVE-NG lab environment implementing FortiGate 7.2.0 with High Availability, dual-site IPsec VPN, SD-WAN, authentication, and security profiles. It is intended for network engineers, security architects, and Fortinet professionals seeking a structured training and reference environment.

1.2 Project Scope

The following technologies and configurations are covered in this lab:

- Active-Passive HA clusters at both HQ and BR sites (FG1-HQ / FG2-HQ, FG1-BR / FG2-BR)
- Redundant site-to-site IPsec VPN tunnels using IKEv2
- SD-WAN with dual ISP connectivity and VPN overlay zones
- Local firewall authentication with trusted host restrictions
- Flow-based Antivirus (LAB-AV) and Web Filter (LAB-WEB) security profiles
- Full verification, troubleshooting, and maintenance procedures

1.3 Key Objectives

The primary objectives of this lab implementation are:

- Provide a reproducible multi-site FortiGate lab on EVE-NG
- Demonstrate enterprise-grade HA and redundant WAN connectivity
- Validate SD-WAN failover between WAN underlay and VPN overlay paths
- Test security profile enforcement at the policy level
- Serve as a hands-on training reference for Fortinet NSE certification preparation

1.4 Technologies Implemented

Technology	Version / Standard	Scope
FortiOS	7.2.0	All FortiGate devices
High Availability	Active-Passive	HQ and BR clusters
IPsec VPN	IKEv2, AES256-SHA256	Site-to-Site HQ ↔ BR
SD-WAN	FortiOS SD-WAN	Dual WAN + VPN overlay
Authentication	Local DB, Firewall Auth	Web policy authentication
Antivirus	Flow-based (LAB-AV)	HTTP / FTP scanning
Web Filter	URL filter (LAB-WEB)	blocked.lab enforcement
EVE-NG	Pro / Community	Lab virtualisation platform

2. Lab Topology Overview

2.1 Physical Topology Diagram

The diagram below shows the complete EVE-NG lab topology including ISP routers, aggregation switches, FortiGate HA pairs, core/distribution switches, and end devices at both the HQ and Branch (BR) sites.

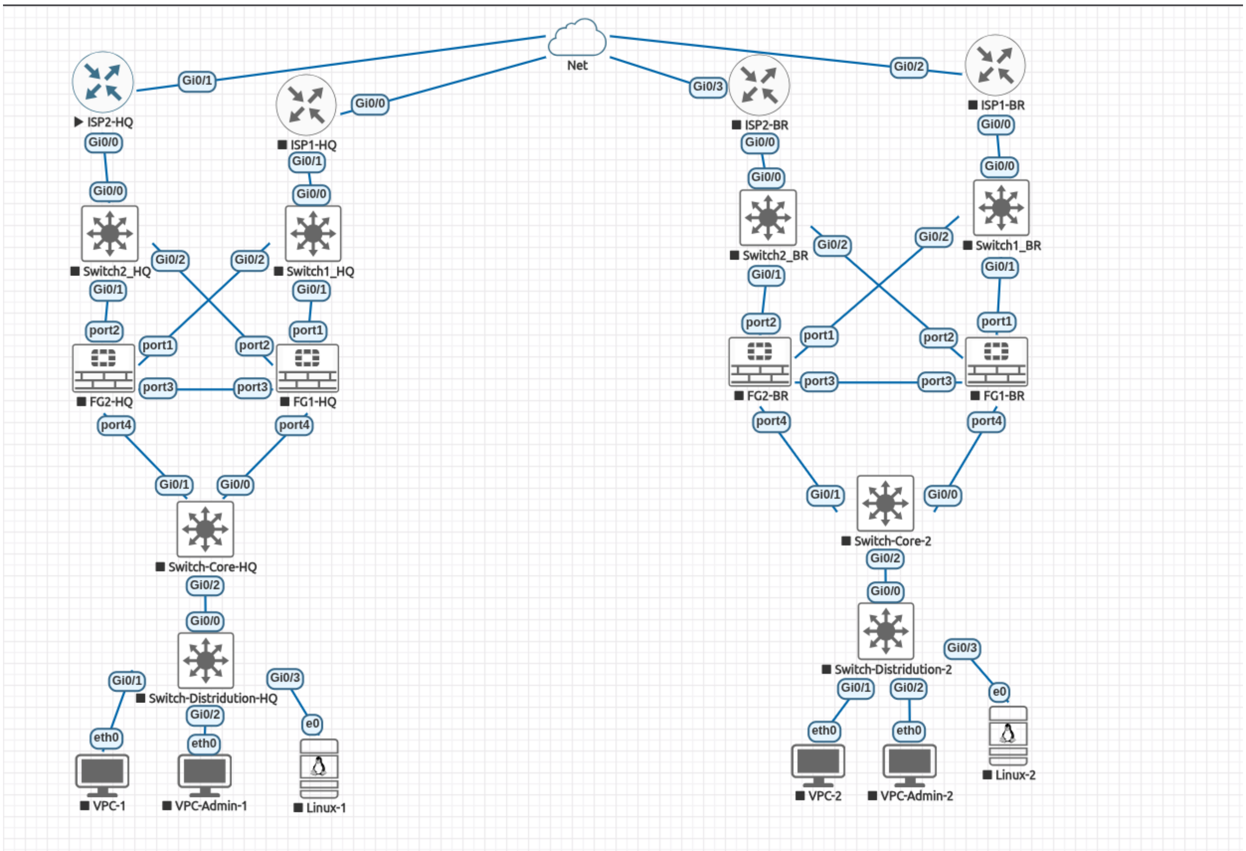


Figure 1 – EVE-NG Lab Topology (HQ & Branch)

2.2 Logical Network Diagram

Key logical relationships in this topology:

- HQ site: ISP1-HQ and ISP2-HQ connect through Switch1_HQ / Switch2_HQ to FG1-HQ (primary) and FG2-HQ (secondary)
- BR site: ISP1-BR and ISP2-BR connect through Switch1_BR / Switch2_BR to FG1-BR (primary) and FG2-BR (secondary)
- Both HA pairs share port3 for heartbeat; port4 connects to the internal LAN via Switch-Core
- Switch-Core-HQ → Switch-Distridution-HQ → VPC-1, VPC-Admin-1, Linux-1
- Switch-Core-2 → Switch-Distridution-2 → VPC-2, VPC-Admin-2, Linux-2
- All ISP routers uplink through a shared cloud/Net segment in EVE-NG

2.3 IP Addressing Plan

Site	Network	Subnet Mask	Gateway	VLAN	Purpose
HQ	192.168.10.0/2	/24	192.168.10.1	10	HQ LAN

Site	Network	Subnet Mask	Gateway	VLAN	Purpose
	4				
HQ	172.16.11.0/30	/30	172.16.11.2	11	HQ ISP1 WAN (port1)
HQ	172.16.12.0/30	/30	172.16.12.2	12	HQ ISP2 WAN (port2)
HQ	10.255.0.0/24	/24	10.255.0.11	-	EVE-NG Management Net
BR	192.168.20.0/24	/24	192.168.20.1	20	BR LAN
BR	172.16.21.0/30	/30	172.16.21.2	21	BR ISP1 WAN (port1)
BR	172.16.22.0/30	/30	172.16.22.2	22	BR ISP2 WAN (port2)

2.4 VLAN Architecture

VLAN ID	Name	Site	Devices
10	HQ_LAN	HQ	FG1-HQ port4, FG2-HQ port4, Switch-Core-HQ
11	HQ_ISP1	HQ	ISP1-HQ Gi0/1, Switch1_HQ, FG1-HQ port1, FG2-HQ port1
12	HQ_ISP2	HQ	ISP2-HQ Gi0/0, Switch2_HQ, FG1-HQ port2, FG2-HQ port2
20	BR_LAN	BR	FG1-BR port4, FG2-BR port4, Switch-Core-2
21	BR_ISP1	BR	ISP1-BR Gi0/0, Switch1_BR, FG1-BR port1, FG2-BR port1
22	BR_ISP2	BR	ISP2-BR Gi0/0, Switch2_BR, FG1-BR port2, FG2-BR port2

3. Hardware & Software Requirements

3.1 EVE-NG Platform Specifications

Component	Minimum	Recommended
CPU	Intel/AMD with VT-x/AMD-V	Intel Xeon / AMD EPYC
RAM	32 GB	64 – 128 GB
Storage	250 GB SSD	500 GB NVMe SSD
Network	Internal bridge for mgmt	Dedicated NIC per segment
OS	Ubuntu 20.04	EVE-NG Pro Edition

3.2 Virtual Machine Requirements

Device Type	Image Required	Count	RAM / vCPU
FortiGate VM	FortiGate-VM64-KVM.vmdk (FortiOS 7.2.0)	4	2 GB / 2 vCPU
Cisco Router	Cisco IOSv 15.x	4	512 MB / 1 vCPU
Cisco L2 Switch	Cisco IOSvL2 15.x	6	512 MB / 1 vCPU
Linux Endpoint	Slax Linux (latest)	2	256 MB / 1 vCPU
VPC	VPCS (latest)	4	64 MB / 1 vCPU

3.3 Software Versions

- FortiOS: 7.2.0 (GA release)
- Cisco IOS: 15.x for both router and L2 switch images
- EVE-NG: Community or Pro edition (2.0.3-112 or later)
- Linux: Slax or any lightweight distro supporting wget, curl, Python 3

3.4 License Requirements

- FortiGate VM: Evaluation license or valid subscription (FortiCare + UTM bundle required for AV/WebFilter)

- EVE-NG Pro license required for more than 9 nodes (Community edition is sufficient for this lab)

4. Pre-Configuration Checklist

4.1 Clean Lab Verification

☐ EVE-NG instance is running and reachable
☐ All required VM images are uploaded to EVE-NG image store
☐ All FortiGates are factory-reset (no previous config)
☐ All network connections are wired per the topology diagram
☐ Administrative access confirmed via console on each device

4.2 Image Verification

☐ FortiGate-VM64-KVM.vmdk (FortiOS 7.2.0) present in /opt/unetlab/addons/qemu/
☐ Cisco IOSv image present
☐ Cisco IOSvL2 image present
☐ VPCS and Slax Linux images present

4.3 Connectivity Pre-Checks

☐ ISP routers reach the EVE-NG cloud Net segment
☐ Ping between ISP1-HQ Gi0/0 and EVE-Net (10.255.0.x)
☐ Console access to all FortiGates verified
☐ VPCs can reach default gateway after FortiGate LAN interface is up

4.4 Environment Variables

Replace the following placeholder values throughout all configurations before deployment:

Placeholder	Description	Example Value
REPLACE_HQ_HA_PASSWORD	HA group password for HQ cluster	Str0ngH@Pass!
REPLACE_BR_HA_PASSWORD	HA group password for BR cluster	Br@nch#HA99
REPLACE_PSK_WAN1	IPsec PSK for WAN1 tunnels (HQ-BR-WAN1 / BR-HQ-WAN1)	W@n1Psk2024!
REPLACE_PSK_WAN2	IPsec PSK for WAN2 tunnels (HQ-BR-WAN2 / BR-HQ-WAN2)	W@n2Psk2024!
REPLACE_ADMIN_PASS	Local admin account password	L@bAdm1n!
REPLACE_USER_PASS	Firewall user (hq-user/br-user) password	U$er@2024

5. Infrastructure Configuration

5.1 HQ Switches

Switch1_HQ – ISP1 Aggregation

Purpose: Connects ISP1-HQ to both HQ FortiGate units (port1) for WAN1 redundancy.

Port	Purpose	Mode	VLAN
Gi0/0	TO_ISP1_HQ	Access	11
Gi0/1	TO_FG1_HQ port1	Access	11
Gi0/2	TO_FG2_HQ port1	Access	11

```
hostname Switch1_HQ
!
vlan 11
 name HQ_ISP1
!
interface GigabitEthernet0/0
 description TO_ISP1_HQ
 switchport mode access
 switchport access vlan 11
 no shutdown
!
interface GigabitEthernet0/1
 description TO_FG1_HQ_PORT1
 switchport mode access
 switchport access vlan 11
 no shutdown
!
interface GigabitEthernet0/2
 description TO_FG2_HQ_PORT1
 switchport mode access
 switchport access vlan 11
 no shutdown
```

Switch2_HQ – ISP2 Aggregation

Purpose: Connects ISP2-HQ to both HQ FortiGate units (port2) for WAN2 redundancy.

Port	Purpose	Mode	VLAN
Gi0/0	TO_ISP2_HQ	Access	12
Gi0/1	TO_FG1_HQ port2	Access	12
Gi0/2	TO_FG2_HQ port2	Access	12

Switch-Core-HQ – LAN Core

Purpose: Connects both HQ FortiGate port4 interfaces to internal distribution layer.

Port	Purpose	Mode	VLAN
Gi0/0	TO_FG1_HQ port4	Access	10
Gi0/1	TO_FG2_HQ port4	Access	10
Gi0/2	TO_Switch-Distridution-HQ	Access	10

5.2 BR Switches

Branch switches follow identical logic to HQ switches with VLAN IDs 21 (ISP1), 22 (ISP2), and 20 (LAN):

- Switch1_BR: Connects ISP1-BR to FG1-BR port1 and FG2-BR port1 (VLAN 21)
- Switch2_BR: Connects ISP2-BR to FG1-BR port2 and FG2-BR port2 (VLAN 22)
- Switch-Core-2: Connects FG1-BR port4 and FG2-BR port4 to Switch-Distridution-2 (VLAN 20)

5.3 ISP Routers

Router	Gi0/0 (Net)	Gi0/1 (WAN)	ISP Network
ISP1-HQ	10.255.0.11/24	172.16.11.1/30	172.16.11.0/30
ISP2-HQ	10.255.0.12/24	172.16.12.1/30	172.16.12.0/30
ISP1-BR	10.255.0.21/24	172.16.21.1/30	172.16.21.0/30
ISP2-BR	10.255.0.22/24	172.16.22.1/30	172.16.22.0/30

6. FortiGate HA Configuration

6.1 HQ HA Cluster Setup

FG1-HQ – Primary (Priority 200)

```
config system global
    set hostname "FG1-HQ"
end

config system ha
    set group-id 10
    set group-name "HQ-HA"
    set mode a-p
    set password "REPLACE_HQ_HA_PASSWORD"
    set hbdev "port3" 100
    set session-pickup enable
    set session-pickup-connectionless enable
    set monitor "port4"
    set override enable
    set priority 200
end
```

FG2-HQ – Secondary (Priority 100)

```
config system global
    set hostname "FG2-HQ"
end

config system ha
    set group-id 10
    set group-name "HQ-HA"
    set mode a-p
    set password "REPLACE_HQ_HA_PASSWORD"
    set hbdev "port3" 100
    set session-pickup enable
    set session-pickup-connectionless enable
    set monitor "port4"
    set override enable
    set priority 100
end
```

6.2 BR HA Cluster Setup

FG1-BR – Primary (Priority 200)

```
config system global
    set hostname "FG1-BR"
end

config system ha
    set group-id 20
    set group-name "BR-HA"
    set mode a-p
    set password "REPLACE_BR_HA_PASSWORD"
    set hbdev "port3" 100
    set session-pickup enable
    set session-pickup-connectionless enable
    set monitor "port4"
    set override enable
```

```
    set priority 200
end
```

FG2-BR – Secondary (Priority 100)

```
config system global
    set hostname "FG2-BR"
end

config system ha
    set group-id 20
    set group-name "BR-HA"
    set mode a-p
    set password "REPLACE_BR_HA_PASSWORD"
    set hbdev "port3" 100
    set session-pickup enable
    set monitor "port4"
    set override enable
    set priority 100
end
```

6.3 HA Verification Commands

```
get system ha status
diagnose sys ha checksum cluster
diagnose sys ha dump-by group
```

Expected: Both units show the same checksum. FG1 shows "Primary" role; FG2 shows "Secondary".

7. FortiGate Network Configuration

7.1 Interface Configuration

HQ Interface Table

Interface	Alias	IP Address	Mask	Role	Description
port1	HQ_ISP1	172.16.11.2	/30	WAN	WAN1 → ISP1-HQ
port2	HQ_ISP2	172.16.12.2	/30	WAN	WAN2 → ISP2-HQ
port3	HQ_HA	N/A	N/A	HA	HA heartbeat
port4	HQ_LAN	192.168.10.1	/24	LAN	Inside LAN gateway

BR Interface Table

Interface	Alias	IP Address	Mask	Role	Description
port1	BR_ISP1	172.16.21.2	/30	WAN	WAN1 → ISP1-BR
port2	BR_ISP2	172.16.22.2	/30	WAN	WAN2 → ISP2-BR
port3	BR_HA	N/A	N/A	HA	HA heartbeat
port4	BR_LAN	192.168.20.1	/24	LAN	Inside LAN gateway

HQ Interface Configuration (CLI)

```
config system interface
    edit "port1"
        set alias "HQ_ISP1"
        set mode static
        set ip 172.16.11.2 255.255.255.252
        set allowaccess ping
        set role wan
    next
    edit "port2"
        set alias "HQ_ISP2"
        set mode static
        set ip 172.16.12.2 255.255.255.252
```

```
        set allowaccess ping
        set role wan
    next
    edit "port4"
        set alias "HQ_LAN"
        set mode static
        set ip 192.168.10.1 255.255.255.0
        set allowaccess ping https ssh http
        set role lan
    next
end
```

7.2 Account Configuration

Admin account trusted-host restrictions limit GUI/CLI access to authorised management workstations:

Site	Admin Account	Trusted Host	Management Workstation
HQ	hq-admin	192.168.10.20/32	Linux-1 (eth0)
BR	br-admin	192.168.20.20/32	Linux-2 (eth0)

```
# HQ – create restricted admin
config system admin
    edit "hq-admin"
        set password "REPLACE_ADMIN_PASS"
        set trusthost1 192.168.10.20 255.255.255.255
        set accprofile "super_admin"
    next
    # Restrict built-in admin AFTER testing hq-admin
    edit "admin"
        set trusthost1 192.168.10.20 255.255.255.255
    next
end
```

7.3 Firewall Address Objects

Object Name	Type	Subnet / FQDN	Site
HQ-LAN	Subnet	192.168.10.0/24	HQ
BR-LAN	Subnet	192.168.20.0/24	BR
allowed.lab	FQDN	allowed.lab	HQ/BR
blocked.lab	FQDN	blocked.lab	HQ/BR

8. Security Profiles

8.1 Antivirus Profile – LAB-AV

A flow-based antivirus profile applied to outbound/inter-site HTTP and FTP traffic for malware detection.

```
config antivirus profile
    edit "LAB-AV"
        set comment "Flow antivirus for EVE lab"
        set feature-set flow
        config http
            set av-scan block
        end
        config ftp
            set av-scan block
        end
    next
end
```

Test: Download EICAR test file from Linux-1 via wget; the connection should be blocked and logged.

8.2 Web Filter Profile – LAB-WEB

A URL filter profile that blocks access to the domain blocked.lab while permitting allowed.lab.

```
config webfilter urlfilter
```

```
    edit 100
        set name "HQ-URL-FILTER"
        config entries
            edit 1
                set url "blocked.lab"
                set type simple
                set action block
            next
        end
    next
end

config webfilter profile
    edit "LAB-WEB"
        set comment "Web filter for EVE lab"
        set feature-set flow
        config web
            set urlfilter-table 100
        end
    next
end
```

8.3 Profile Verification

- AV: confirmed via block of EICAR file download over HTTP
- WebFilter: confirmed via block page when accessing http://blocked.lab/ from Linux-1
- Log inspection: Security logs should show threat name and action=blocked

9. VPN Configuration

9.1 Tunnel Overview

Tunnel Name	Site	Interface	Remote Gateway	PSK Placeholder
HQ-BR-WAN1	HQ	port1	172.16.21.2	REPLACE_PSK_WAN1
HQ-BR-WAN2	HQ	port2	172.16.22.2	REPLACE_PSK_WAN2
BR-HQ-WAN1	BR	port1	172.16.11.2	REPLACE_PSK_WAN1
BR-HQ-WAN2	BR	port2	172.16.12.2	REPLACE_PSK_WAN2

9.2 Phase 1 Configuration (HQ-BR-WAN1 example)

```
config vpn ipsec phase1-interface
    edit "HQ-BR-WAN1"
        set interface "port1"
        set ike-version 2
        set keylife 28800
        set peertype any
        set proposal aes256-sha256
        set dhgrp 14
        set remote-gw 172.16.21.2
        set psksecret "REPLACE_PSK_WAN1"
        set dpd on-idle
        set dpd-retrycount 3
        set dpd-retryinterval 5
    next
end
```

9.3 Phase 2 Configuration (HQ-BR-WAN1 example)

```
config vpn ipsec phase2-interface
    edit "HQ-BR-WAN1"
        set phase1name "HQ-BR-WAN1"
        set proposal aes256-sha256
        set dhgrp 14
        set pfs enable
```

```
        set keylifeseconds 3600
        set auto-negotiate enable
    next
end
```

Repeat Phase 1 and Phase 2 configurations for HQ-BR-WAN2, BR-HQ-WAN1, and BR-HQ-WAN2 with their respective interfaces, remote gateways, and PSKs.

10. SD-WAN Configuration

10.1 SD-WAN Zones

Zone Name	Type	Members
WAN-UNDERLAY	WAN	port1, port2
VPN-OVERLAY	Overlay	HQ-BR-WAN1, HQ-BR-WAN2

10.2 SD-WAN Members

ID	Interface	Zone	Gateway	Role
1	port1	WAN-UNDERLAY	172.16.11.1	Primary WAN
2	port2	WAN-UNDERLAY	172.16.12.1	Secondary WAN
11	HQ-BR-WAN1	VPN-OVERLAY	N/A (tunnel)	Primary VPN
12	HQ-BR-WAN2	VPN-OVERLAY	N/A (tunnel)	Secondary VPN

10.3 Health Checks

Name	Destination	Protocol	Members	Latency	Loss
HQ-WAN1-E2E	172.16.21.2	Ping	1	200 ms	20 %
HQ-WAN2-E2E	172.16.22.2	Ping	2	200 ms	20 %
HQ-VPN-E2E	192.168.20.1	Ping	11, 12	200 ms	20 %

10.4 SD-WAN Services

ID	Name	Source	Destination	Priority Order
10	HQ-TO-BR-VPN	HQ-LAN	BR-LAN	11, 12
20	HQ-WAN-DEFAULT	HQ-LAN	all	1, 2

10.5 SD-WAN CLI Configuration Snippet

```
config system sdwan
    set status enable
    config zone
        edit "WAN-UNDERLAY"
        next
        edit "VPN-OVERLAY"
        next
    end
    config members
        edit 1
            set interface "port1"
            set zone "WAN-UNDERLAY"
            set gateway 172.16.11.1
        next
        edit 2
            set interface "port2"
            set zone "WAN-UNDERLAY"
            set gateway 172.16.12.1
        next
        edit 11
            set interface "HQ-BR-WAN1"
```

```
            set zone "VPN-OVERLAY"
        next
        edit 12
            set interface "HQ-BR-WAN2"
            set zone "VPN-OVERLAY"
        next
    end
end
```

11. Firewall Policies

11.1 HQ Policies

Policy 10 – HQ-BR-PING

Parameter	Value
Name	HQ-BR-PING
Source Interface	port4 (HQ_LAN)
Destination Interface	VPN-OVERLAY
Source Address	HQ-LAN
Destination Address	BR-LAN
Action	Accept
Service	PING
NAT	Disabled
Logging	All Sessions

Policy 20 – HQ-BR-WEB-AUTH-UTM

Parameter	Value
Name	HQ-BR-WEB-AUTH-UTM
Source Interface	port4 (HQ_LAN)
Destination Interface	VPN-OVERLAY
Source Address	HQ-LAN
Destination Address	BR-LAN
Action	Accept
Service	HTTP, HTTPS
Authentication	Firewall auth (hq-user)
Antivirus Profile	LAB-AV
Web Filter Profile	LAB-WEB
NAT	Disabled

Policy 30 – HQ-WAN-DIAGNOSTIC

Parameter	Value
Name	HQ-WAN-DIAGNOSTIC
Source Interface	port4 (HQ_LAN)
Destination Interface	WAN-UNDERLAY
Source Address	HQ-LAN
Destination Address	all
Action	Accept
Service	PING, DNS, HTTP
NAT	Enabled (masquerade)

11.2 BR Policies

BR policies mirror the HQ configuration with BR-specific address objects and interfaces:

- Policy 10: BR-HQ-PING — port4 → VPN-OVERLAY, BR-LAN → HQ-LAN, PING, Accept
- Policy 20: BR-HQ-WEB-AUTH-UTM — port4 → VPN-OVERLAY, HTTP/HTTPS, auth + UTM
- Policy 30: BR-WAN-DIAGNOSTIC — port4 → WAN-UNDERLAY, diagnostic traffic, NAT enabled

12. Verification Procedures

12.1 HA Verification

Check	HQ Expected	BR Expected
FG1 Role	Primary	Primary
FG2 Role	Secondary	Secondary
HA Group	HQ-HA	BR-HA
HA Mode	Active-Passive	Active-Passive
Heartbeat Interface	port3	port3
Monitored Interface	port4	port4
Checksum	Both units match	Both units match

```
get system ha status
diagnose sys ha checksum cluster
diagnose sys ha dump-by group
```

12.2 Interface Verification

Interface	IP	Expected Status	Connectivity Test
port1 (HQ)	172.16.11.2	Up	Ping 172.16.11.1 (ISP1-HQ)
port2 (HQ)	172.16.12.2	Up	Ping 172.16.12.1 (ISP2-HQ)
port4 (HQ)	192.168.10.1	Up	Ping from VPC-1 / Linux-1
port1 (BR)	172.16.21.2	Up	Ping 172.16.21.1 (ISP1-BR)
port2 (BR)	172.16.22.2	Up	Ping 172.16.22.1 (ISP2-BR)
port4 (BR)	192.168.20.1	Up	Ping from VPC-2 / Linux-2

12.3 VPN Verification

```
get vpn ipsec tunnel summary
diagnose vpn tunnel list
diagnose vpn ike log-filter list
```

Expected: All 4 tunnels (WAN1 and WAN2 for both directions) show Phase 1 and Phase 2 as Established.

12.4 SD-WAN Verification

```
diagnose sys sdwan member
diagnose sys sdwan health-check
diagnose sys sdwan service
diagnose sys sdwan intf-sla-log
```

13. Testing Scenarios

13.1 HA Failover Test

Step	Action	Expected Result
1	Verify FG1 is primary via get system ha status	FG1 shows "Primary"
2	Initiate continuous ping from VPC-1 to VPC-2	Ping succeeds
3	Shutdown port4 on FG1-HQ (cable unplug / shutdown)	FG2 becomes primary within ~10s
4	Check HA status on FG2	FG2 now shows "Primary"
5	Restore port4 on FG1	FG1 pre-empts and returns to primary (override=enable)
6	Confirm ping continuity	Minimal packet loss (<5 packets)

13.2 VPN Connectivity Test

Step	Action	Expected Result
1	Verify both VPN tunnels are up	Phase 1 and 2 established for WAN1 and WAN2
2	Ping from VPC-1 (192.168.10.x) to VPC-2 (192.168.20.x)	ICMP replies received
3	Check SD-WAN route for 192.168.20.0/24	Route via VPN-OVERLAY
4	Simulate WAN1 failure; check SD-WAN failover to WAN2	Traffic continues over WAN2 VPN

13.3 Authentication Test

Step	Action	Expected Result
1	Start HTTP server on Linux-2: python3 -m http.server 80	Server listening on port 80
2	From Linux-1, add hosts entry: 192.168.20.20 allowed.lab	Name resolves to Linux-2
3	Open http://allowed.lab/ in Linux-1 browser	FortiGate shows auth portal
4	Enter hq-user credentials	Login succeeds; page loads
5	Run: diagnose firewall auth list	hq-user shows as authenticated
6	Access http://blocked.lab/	Web filter block page displayed

13.4 SD-WAN Failover Test

Step	Action	Expected Result
1	Run continuous ping from Linux-1 to Linux-2	Ping succeeds via VPN-OVERLAY
2	Shutdown ISP1-HQ Gi0/1 (WAN1 link)	Health check for WAN1 fails
3	Check SD-WAN health check status	WAN1 member shows SLA fail
4	Verify traffic moves to WAN2 VPN	Ping continues via HQ-BR-WAN2
5	Restore WAN1 link	Traffic returns to WAN1 (primary)

14. Troubleshooting Guide

14.1 Common Issues Matrix

Issue	Symptoms	Root Cause	Resolution
HA not syncing	Checksum mismatch	Config difference between units	Compare configs; manually resync
VPN not establishing	Tunnel stays down	PSK mismatch or routing error	Verify PSK; check phase1/2 settings
SD-WAN failover fails	Traffic stays on failed path	Health check SLA not triggered	Check health-check reachability
Auth not prompting	Users pass without auth	Policy mis-order; auth not enabled	Move auth policy above accept policy
AV not blocking	EICAR download succeeds	Profile not attached to policy	Verify UTM profile in firewall policy

14.2 Flow Debug Procedure

```
# Clear previous debug state
diagnose debug reset
diagnose debug flow filter clear

# Set filters for specific traffic
diagnose debug flow filter saddr 192.168.10.100
diagnose debug flow filter daddr 192.168.20.100

# Enable debug output
diagnose debug flow show function-name enable
diagnose debug flow trace start 50
diagnose debug enable

# Reproduce the traffic, then stop debug
diagnose debug disable
diagnose debug reset
```

14.3 IKE Debug (VPN)

```
diagnose vpn ike log-filter clear
diagnose vpn ike log-filter name HQ-BR-WAN1
diagnose debug application ike -1
diagnose debug enable

# To stop:
diagnose debug disable
diagnose debug reset
```

14.4 Authentication Debug

```
diagnose firewall auth list
diagnose firewall auth clear
diagnose debug application fnbamd -1
diagnose debug enable
```

15. Maintenance Procedures

15.1 Backup Configuration

```
# Backup HQ configuration via TFTP
execute backup config tftp backup_hq.conf 192.168.10.20

# Backup BR configuration via TFTP
execute backup config tftp backup_br.conf 192.168.20.20

# Display running config (copy-paste to file)
show full-configuration
```

15.2 Monitoring Guidelines

Key metrics to monitor on a regular basis:

```
# CPU and Memory
get system performance status

# VPN Tunnel Status
get vpn ipsec tunnel summary

# SD-WAN Health
diagnose sys sdwan health-check

# HA Synchronisation
diagnose sys ha checksum cluster

# Security / Auth
diagnose firewall auth list
diagnose autoupdate versions
```

15.3 Firmware Upgrade Procedure

- Download FortiOS firmware from support.fortinet.com
- Upload firmware via GUI: System → Firmware → Upload
- For HA cluster: perform upgrade on secondary first, then primary
- Verify both units re-join cluster post-upgrade
- Validate all policies, VPN tunnels, and SD-WAN services after upgrade

16. Appendices

16.1 Full Troubleshooting Command Reference

High Availability

```
get system ha status
diagnose sys ha checksum cluster
diagnose sys ha dump-by group
diagnose sys ha reset-uptime
```

VPN

```
get vpn ipsec tunnel summary
diagnose vpn tunnel list
diagnose vpn ike log-filter list
execute vpn ipsec restart <tunnel-name>
```

SD-WAN

```
diagnose sys sdwan member
diagnose sys sdwan health-check
diagnose sys sdwan service
diagnose sys sdwan intf-sla-log
diagnose sys sdwan internet-service-ctrl list
```

Security Profiles

```
diagnose firewall auth list
diagnose firewall auth clear
diagnose autoupdate versions
diagnose ips session list
```

16.2 Change Log

Date	Version	Author	Description
2024-XX-XX	1.0	Lab Team	Initial documentation created
2024-XX-XX	1.1	Lab Team	Added verification and troubleshooting procedures
2024-XX-XX	1.2	Lab Team	Topology diagram and testing scenarios added

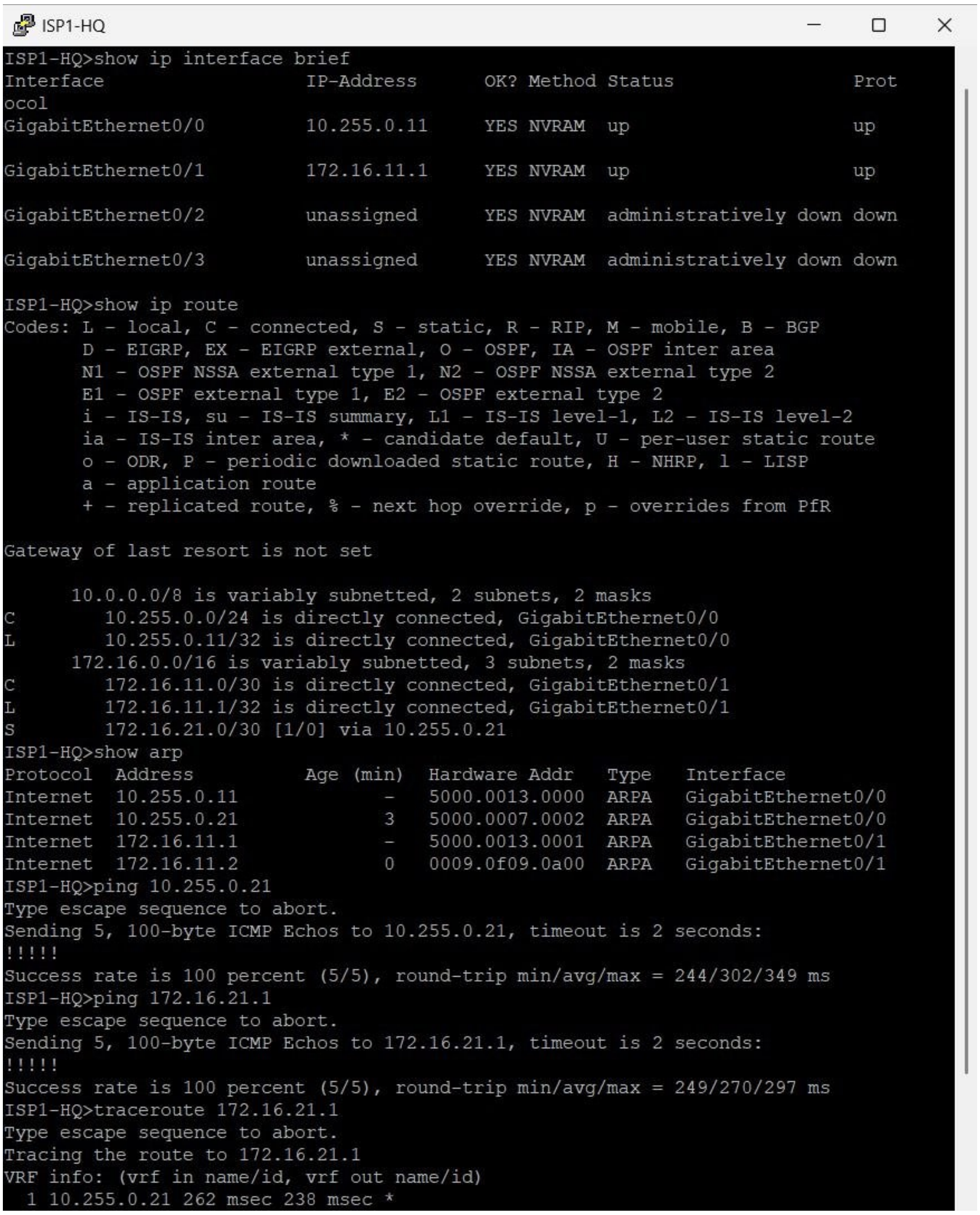
17. Lab Evidence – Proof of Work

This section contains verified output screenshots captured directly from EVE-NG console sessions. They confirm that the lab infrastructure (ISP routers, HQ LAN, end devices) is correctly configured and reachable. Note: pings to the Branch (192.168.20.x) return "Destination network unreachable" at this stage — this is expected behaviour because the IPsec VPN tunnels and SD-WAN overlay routes have not yet been applied. The error originates from the FortiGate (192.168.10.1), which has no route to the BR LAN, confirming the firewall is the next-hop but the VPN is pending configuration.

17.1 ISP1-HQ – Interface & Connectivity Verification

Device: ISP1-HQ | Gi0/0 → 10.255.0.11 (EVE-Net) | Gi0/1 → 172.16.11.1 (WAN1 toward HQ FortiGates).

- Static route to 172.16.21.0/30 (BR WAN1 subnet) via 10.255.0.21 (ISP1-BR) is installed.
- Ping to 10.255.0.21 (ISP1-BR Gi0/0) → 100% success confirms EVE-Net reachability.
- Ping to 172.16.21.1 (BR WAN1 ISP-side IP) → 100% success confirms end-to-end ISP path.
- ARP table shows 172.16.11.2 (FG1-HQ port1) as active neighbour on Gi0/1.



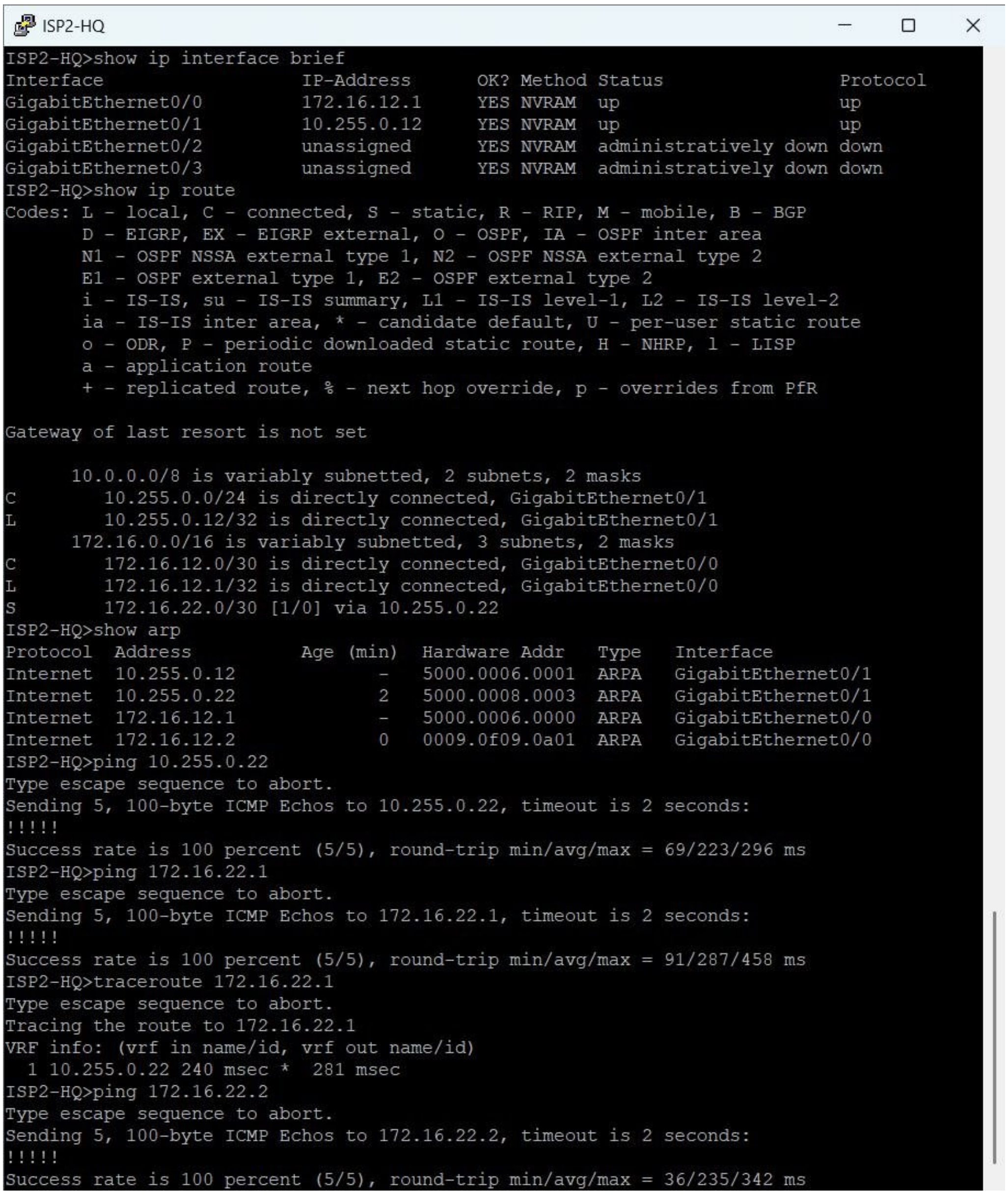
```
ISP1-HQ
ISP1-HQ>show ip interface brief
Interface                  IP-Address      OK? Method Status                Prot
ocol
GigabitEthernet0/0         10.255.0.11     YES NVRAM  up                    up

GigabitEthernet0/1         172.16.11.1     YES NVRAM  up                    up

GigabitEthernet0/2         unassigned      YES NVRAM  administratively down down

GigabitEthernet0/3         unassigned      YES NVRAM  administratively down down

ISP1-HQ>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.255.0.0/24 is directly connected, GigabitEthernet0/0
L        10.255.0.11/32 is directly connected, GigabitEthernet0/0
      172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
C        172.16.11.0/30 is directly connected, GigabitEthernet0/1
L        172.16.11.1/32 is directly connected, GigabitEthernet0/1
S        172.16.21.0/30 [1/0] via 10.255.0.21
ISP1-HQ>show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  10.255.0.11             -   5000.0013.0000  ARPA   GigabitEthernet0/0
Internet  10.255.0.21             3   5000.0007.0002  ARPA   GigabitEthernet0/0
Internet  172.16.11.1             -   5000.0013.0001  ARPA   GigabitEthernet0/1
Internet  172.16.11.2             0   0009.0f09.0a00  ARPA   GigabitEthernet0/1
ISP1-HQ>ping 10.255.0.21
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.255.0.21, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 244/302/349 ms
ISP1-HQ>ping 172.16.21.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.21.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 249/270/297 ms
ISP1-HQ>traceroute 172.16.21.1
Type escape sequence to abort.
Tracing the route to 172.16.21.1
VRF info: (vrf in name/id, vrf out name/id)
  1 10.255.0.21 262 msec 238 msec *
```

Figure 2 – ISP1-HQ: show ip interface brief / show ip route / show arp / ping verification

17.2 ISP2-HQ – Interface & Connectivity Verification

Device: ISP2-HQ | Gi0/0 → 172.16.12.1 (WAN2 toward HQ FortiGates) | Gi0/1 → 10.255.0.12 (EVE-Net).

- Static route to 172.16.22.0/30 (BR WAN2 subnet) via 10.255.0.22 (ISP2-BR) is installed.
- Ping to 10.255.0.22 (ISP2-BR) → 100% success; traceroute confirms single-hop via EVE-Net.
- Ping to 172.16.22.1 (BR WAN2 ISP-side IP) → 100% success confirms WAN2 path end-to-end.
- Ping to 172.16.22.2 (FG1-BR port2 WAN2 IP) → 100% success confirms FortiGate WAN2 is live.
- ARP table shows 172.16.12.2 (FG1-HQ port2) as active neighbour — confirms FG WAN2 interface is up.



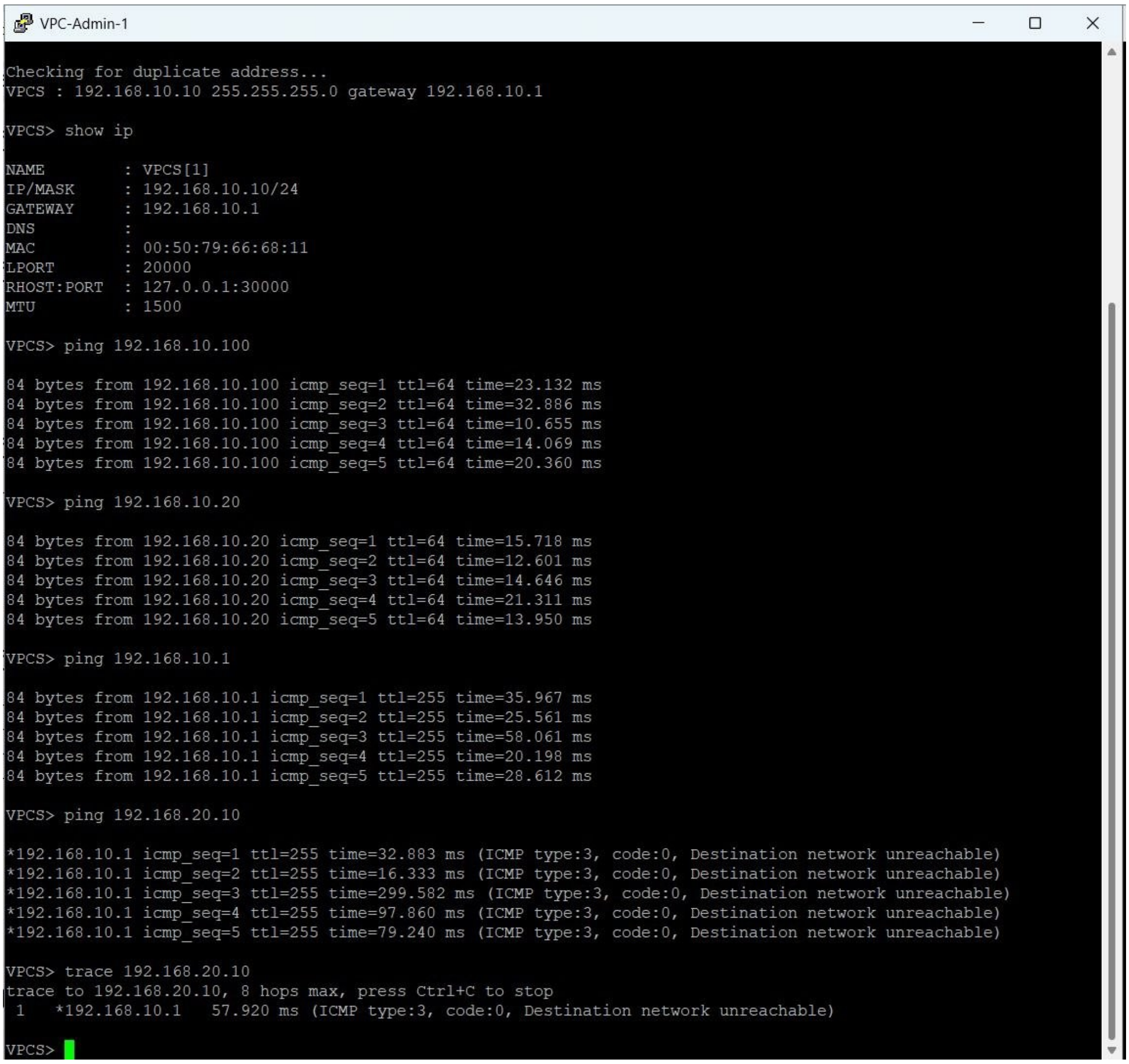
```
ISP2-HQ>show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0         172.16.12.1     YES NVRAM  up                    up
GigabitEthernet0/1         10.255.0.12     YES NVRAM  up                    up
GigabitEthernet0/2         unassigned      YES NVRAM  administratively down down
GigabitEthernet0/3         unassigned      YES NVRAM  administratively down down
ISP2-HQ>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.255.0.0/24 is directly connected, GigabitEthernet0/1
L        10.255.0.12/32 is directly connected, GigabitEthernet0/1
      172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
C        172.16.12.0/30 is directly connected, GigabitEthernet0/0
L        172.16.12.1/32 is directly connected, GigabitEthernet0/0
S        172.16.22.0/30 [1/0] via 10.255.0.22
ISP2-HQ>show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  10.255.0.12             -   5000.0006.0001  ARPA   GigabitEthernet0/1
Internet  10.255.0.22             2   5000.0008.0003  ARPA   GigabitEthernet0/1
Internet  172.16.12.1             -   5000.0006.0000  ARPA   GigabitEthernet0/0
Internet  172.16.12.2             0   0009.0f09.0a01  ARPA   GigabitEthernet0/0
ISP2-HQ>ping 10.255.0.22
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.255.0.22, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 69/223/296 ms
ISP2-HQ>ping 172.16.22.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.22.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 91/287/458 ms
ISP2-HQ>traceroute 172.16.22.1
Type escape sequence to abort.
Tracing the route to 172.16.22.1
VRF info: (vrf in name/id, vrf out name/id)
  1 10.255.0.22 240 msec *  281 msec
ISP2-HQ>ping 172.16.22.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.22.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 36/235/342 ms
```

Figure 3 – ISP2-HQ: show ip interface brief / show ip route / show arp / ping & traceroute verification

17.3 VPC-Admin-1 – HQ LAN Verification

Device: VPC-Admin-1 | IP: 192.168.10.10/24 | Gateway: 192.168.10.1 (FG1-HQ port4 – active HA primary).

- Ping to 192.168.10.100 (VPC-1) → 100% success — HQ LAN east-west connectivity confirmed.
- Ping to 192.168.10.20 (Linux-1 / management host) → 100% success.
- Ping to 192.168.10.1 (FortiGate HQ LAN gateway, TTL=255) → 100% success — FG port4 is up.
- Ping to 192.168.20.10 (BR VPC-Admin-2) → "Destination network unreachable" from 192.168.10.1. EXPECTED: the FortiGate has no route to 192.168.20.0/24 because IPsec VPN tunnels and SD-WAN overlay are not yet configured at this stage.
- Traceroute to 192.168.20.10 → stops at 192.168.10.1, confirming the FG is the exit point but has no onward path (VPN pending).



```
VPC-Admin-1

Checking for duplicate address...
VPCS : 192.168.10.10 255.255.255.0 gateway 192.168.10.1

VPCS> show ip

NAME        : VPCS[1]
IP/MASK     : 192.168.10.10/24
GATEWAY     : 192.168.10.1
DNS         :
MAC         : 00:50:79:66:68:11
LPORT       : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500

VPCS> ping 192.168.10.100

84 bytes from 192.168.10.100 icmp_seq=1 ttl=64 time=23.132 ms
84 bytes from 192.168.10.100 icmp_seq=2 ttl=64 time=32.886 ms
84 bytes from 192.168.10.100 icmp_seq=3 ttl=64 time=10.655 ms
84 bytes from 192.168.10.100 icmp_seq=4 ttl=64 time=14.069 ms
84 bytes from 192.168.10.100 icmp_seq=5 ttl=64 time=20.360 ms

VPCS> ping 192.168.10.20

84 bytes from 192.168.10.20 icmp_seq=1 ttl=64 time=15.718 ms
84 bytes from 192.168.10.20 icmp_seq=2 ttl=64 time=12.601 ms
84 bytes from 192.168.10.20 icmp_seq=3 ttl=64 time=14.646 ms
84 bytes from 192.168.10.20 icmp_seq=4 ttl=64 time=21.311 ms
84 bytes from 192.168.10.20 icmp_seq=5 ttl=64 time=13.950 ms

VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=35.967 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=25.561 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=58.061 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=20.198 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=28.612 ms

VPCS> ping 192.168.20.10

*192.168.10.1 icmp_seq=1 ttl=255 time=32.883 ms (ICMP type:3, code:0, Destination network unreachable)
*192.168.10.1 icmp_seq=2 ttl=255 time=16.333 ms (ICMP type:3, code:0, Destination network unreachable)
*192.168.10.1 icmp_seq=3 ttl=255 time=299.582 ms (ICMP type:3, code:0, Destination network unreachable)
*192.168.10.1 icmp_seq=4 ttl=255 time=97.860 ms (ICMP type:3, code:0, Destination network unreachable)
*192.168.10.1 icmp_seq=5 ttl=255 time=79.240 ms (ICMP type:3, code:0, Destination network unreachable)

VPCS> trace 192.168.20.10
trace to 192.168.20.10, 8 hops max, press Ctrl+C to stop
 1   *192.168.10.1   57.920 ms (ICMP type:3, code:0, Destination network unreachable)

VPCS>
```

Figure 4 – VPC-Admin-1: show ip / ping (HQ LAN success / BR unreachable – VPN not yet configured)

17.4 VPC-1 – HQ LAN Verification

Device: VPC-1 | IP: 192.168.10.100/24 | Gateway: 192.168.10.1 (FG1-HQ port4 – active HA primary).

- Ping to 192.168.10.10 (VPC-Admin-1) → 100% success — confirms HQ LAN east-west connectivity.
- Ping to 192.168.10.20 (Linux-1) → 100% success.
- Ping to 192.168.10.1 (FortiGate HQ LAN gateway, TTL=255) → 100% success — HA primary is active.
- Ping to 192.168.20.100 (BR VPC-2) and 192.168.20.20 (Linux-2) → "Destination network unreachable" from 192.168.10.1. EXPECTED: same condition as VPC-Admin-1 above. The ICMP type:3 code:0 response originates at the FortiGate, proving that the gateway is reachable but lacks a route to the Branch LAN — confirming VPN configuration is the required next step.

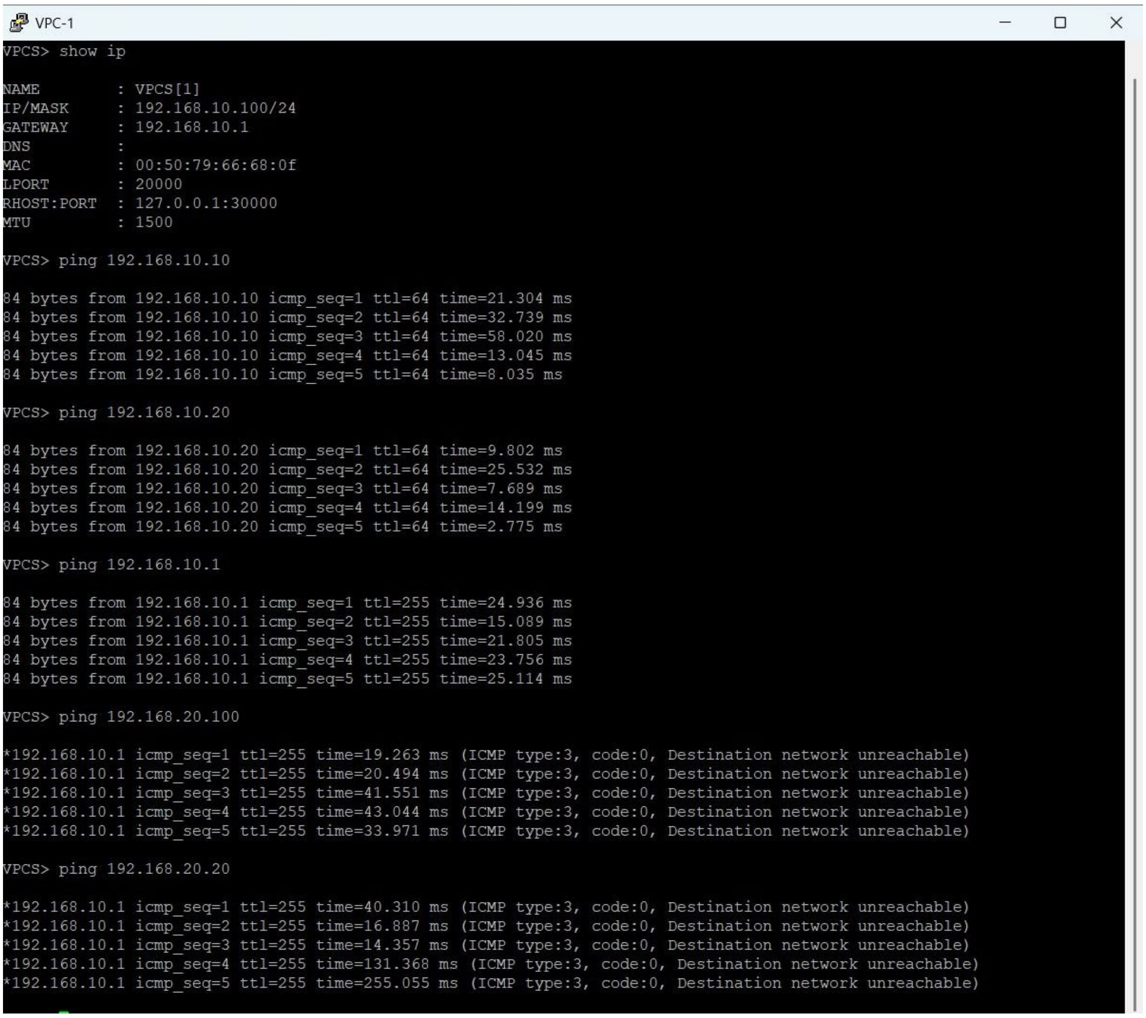


Figure 5 – VPC-1: show ip / ping (HQ LAN success / BR unreachable – VPN not yet configured)

17.5 Connectivity Evidence Summary

Source	Destination	Target IP	Result	Notes
ISP1-HQ	ISP1-BR (EVE-Net)	10.255.0.21	✓ 100% success	EVE-Net transit verified
ISP1-HQ	ISP1-BR WAN	172.16.21.1	✓ 100% success	WAN1 end-to-end path ok
ISP2-HQ	ISP2-BR (EVE-Net)	10.255.0.22	✓ 100% success	EVE-Net transit verified
ISP2-HQ	ISP2-BR WAN	172.16.22.1	✓ 100% success	WAN2 end-to-end path ok
ISP2-HQ	FG1-BR port2	172.16.22.2	✓ 100% success	FortiGate BR WAN2 live
VPC-Admin-1	VPC-1 (HQ LAN)	192.168.10.100	✓ 100% success	HQ LAN east-west ok
VPC-Admin-1	Linux-1 (HQ LAN)	192.168.10.20	✓ 100% success	Management host reachable
VPC-Admin-1	FortiGate HQ GW	192.168.10.1	✓ 100% success	HA primary port4 up
VPC-Admin-1	VPC-Admin-2 (BR)	192.168.20.10	✗ Unreachable	EXPECTED – VPN not configured yet
VPC-1	VPC-Admin-1 (HQ LAN)	192.168.10.10	✓ 100% success	HQ LAN east-west ok
VPC-1	FortiGate HQ GW	192.168.10.1	✓ 100% success	HA primary port4 up
VPC-1	VPC-2 (BR)	192.168.20.100	✗ Unreachable	EXPECTED – VPN not configured yet
VPC-1	Linux-2 (BR)	192.168.20.20	✗ Unreachable	EXPECTED – VPN not configured yet

Key Finding: All HQ-site infrastructure is fully operational. The "Destination network unreachable" responses for Branch (192.168.20.x) addresses are generated by the FortiGate at 192.168.10.1 (ICMP type:3, code:0), not by an intermediate device. This confirms the HQ FortiGate is correctly forwarding and responding, but lacks the VPN tunnel routes to the Branch. Configuring IPsec VPN (Section 9) and SD-WAN overlay (Section 10) will establish this path.